

AGAM TYAGI

agamtyagi2001@gmail.com | +91 82181 85432 | <https://www.linkedin.com/in/agam-tyagi-6624a7204>

SUMMARY

Generative AI-focused Software Engineer with 2+ years of experience building scalable, production-ready applications. Experienced in developing AI-powered solutions using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, vector databases (Qdrant), semantic search, local embedding models, prompt engineering, and AI workflow orchestration. Strong full-stack development expertise with the MERN stack (MongoDB, Express.js, React.js, Node.js), including RESTful APIs, scalable backend systems, authentication, and database design. Passionate about building intelligent, end-to-end applications that combine Generative AI with robust software engineering to solve real-world business problems.

EDUCATION

Master of Computer Applications (MCA) — Chandigarh University	<i>Jul 2022 – Jun 2024</i>
Bachelor of Computer Applications (BCA) — Ch. Charan Singh University	<i>Jul 2019 – Jun 2022</i>

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), PHP

Frontend: React.js, Tailwind CSS, HTML5, CSS3, Bootstrap, jQuery, AJAX

Backend: Node.js, Express.js, FastAPI, RESTful API Development, Laravel, MVC Architecture

Databases: MySQL, MongoDB, PostgreSQL, SQLite, Database Optimization

AI / Generative AI: LLM Integration, RAG (Retrieval-Augmented Generation), LangChain, Prompt Engineering, Semantic Search, Embedding Models, Vector Databases (Qdrant), Document Chunking, Similarity Search, DeepFace, MediaPipe

Auth & Security: JWT, Session Auth, Role-Based Access Control (RBAC), Liveness Detection, Anti-Spoofing

Tools & Platforms: Git, GitHub, Postman, Vercel, Prisma ORM, Mongoose, VS Code, MouseFlow, GA4

Concepts: Data Structures & Algorithms, OOPs, RESTful APIs

WORK EXPERIENCE

Full Stack Developer — Backup Infotech *Jul 2024 – Present*

- Developed a production-ready AI-powered chatbot to automate website query handling using Retrieval-Augmented Generation (RAG).
- Improved chatbot response accuracy by leveraging company-specific knowledge and contextual retrieval, enabling users to receive more relevant and reliable responses.
- Designed and developed production-grade full-stack web applications using React.js, Node.js, Express.js, MongoDB, and REST APIs, implementing secure authentication, dynamic dashboards, advanced search & filtering, and scalable backend architectures across diverse industry domains.
- Built and maintained RESTful APIs consumed by web applications, implementing JWT authentication, RBAC, optimized database queries, and scalable backend architecture following MVC design principles.

PROJECTS

Enterprise AI Chatbot (RAG) *Jun 2026 – Present*

Tech Stack: Node.js, Express.js, LangChain, Groq API, Llama 3.3-70B, Qdrant Vector Database, Transformers.js, all-MiniLM-L6-v2, REST APIs

- Built a production-ready RAG chatbot by crawling and indexing 400+ website pages into a Qdrant vector database for domain-specific knowledge retrieval.
- Implemented lead intent detection and contact information extraction within chatbot APIs, converting conversations into qualified business leads.
- Built a custom retrieval pipeline using document chunking, local embedding generation, vector similarity search, and later migrated it to LangChain for improved maintainability.
- Integrated Groq Llama 3.3-70B with custom retrieval logic, session-based conversation history, and contextual response generation.
- Designed scalable REST APIs for chatbot communication, session management, and seamless integration with frontend applications.

Avencher

Nov 2025 – Present

Tech Stack: React.js (Vite), Node.js, Express.js, MongoDB, AI Integration, REST APIs

- Developed an AI-powered verified talent and hiring platform connecting professionals and companies.
- Implemented secure authentication and identity verification mechanisms for users and employers.
- Designed scalable REST APIs and optimized application performance for handling user and hiring workflows.
- Built AI-driven talent matching and recommendation features to improve hiring efficiency.

Foto-Alpha

Apr 2025 – Dec 2025

Tech Stack: React.js (Vite), Node.js, Express.js, MongoDB, Mongoose, REST APIs

- Developed a full-stack photography management platform enabling photographers to create and manage portfolios, events, and client galleries.
- Implemented QR-based instant gallery delivery and high-speed photo sharing for seamless customer experience.
- Built scalable REST APIs and optimized backend performance for efficient image and gallery management.
- Designed responsive and premium user interfaces with secure cloud storage integration.

Radigone Web App

Oct 2024 – Feb 2025

Tech Stack: PHP, Laravel, HTML/CSS, JavaScript, MySQL

- Developed a multi-role platform (viewers, sponsors, agents) with fine-grained RBAC, permission management, transaction handling, and automated commission calculation.
- Built a comprehensive admin panel for user management, reporting, and platform oversight using Laravel MVC architecture with MySQL as the backend database.

Online Examination System (Internship)

Sep 2024 – Sep 2024

Tech Stack: HTML5, CSS3, Bootstrap, JavaScript, jQuery, AJAX, PHP, Laravel, MySQL

- Built a secure full-featured online exam platform with timed tests, automatic grading, result analytics, and a robust admin panel for exam and question bank management.
- Implemented export functionality for performance reports and used AJAX for dynamic form interactions, ensuring a smooth, responsive examination experience.

PERSONAL PROJECTS

Nexora AI Assistant

Tech Stack: Python, Gradio, Groq API, Ollama, Tavily Search API

- AI-powered conversational assistant with multi-model LLM support (Groq + Ollama) and real-time web search via Tavily API.

AI Resume Parser

Tech Stack: Node.js, Express.js, FastAPI, Python, PDF Parser

- Automated resume parsing system extracting candidate data from PDF/DOCX into structured JSON for ATS ingestion.

AI Face Authentication & Liveness Detection System

Tech Stack: Node.js, Express.js, FastAPI, PostgreSQL, Prisma, DeepFace, MediaPipe

- Developed a biometric authentication system with real-time liveness detection and facial recognition using DeepFace for 128-dimensional face embedding generation and cosine similarity matching. Implemented blink and head-movement verification using MediaPipe to prevent replay and spoofing attacks, enhancing authentication security and accuracy.

Personal Portfolio & Analytics Dashboard

Tech Stack: React.js (Vite), Node.js, Express.js, MongoDB, Mongoose, AI Integration, RAG

- Built a full-stack MERN portfolio with a dynamic admin dashboard enabling real-time content management (profile, skills, experience, education) through custom REST APIs and JWT-based authentication.
- Engineered a custom visitor analytics engine featuring real-time stats, session tracking, and page-level visit recording using cookie-based session middleware and analytics integrations.